Course Guide Outline

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Schedule

- The course ends on **Monday, March 2nd at 23:30 UTC.**
- All course deadlines are listed in Coordinated Universal Time (UTC). Please confirm the alignment to you local time zone.
- Post your questions to the course team in the Discussion Forum.
- Items preceded by a star (✭) are graded with due dates in red below.

Welcome to the Course (35 min) **February 2 — 8, 2026**	
In the first section of the course, you'll take a Pre-Assessment to get a baseline of your understanding of the course material. During this period, you'll become familiar with the platform and course design.	• Entrance Survey (5 min) • ✭ Pre-Assessment (10 min) Due: March 2nd at 23:30 UTC • Welcome (2 min) • Course Discussion Forum (5 min) • Who's in the Course? (3 min) • Who's Teaching the Course (4 min) • LinkedIn Community (3 min)
Week 1: Introduction to Quantum Computing (4-5 hours) **February 2 — 8, 2026**	

The first part of Week 1 is a historical introduction to classical and quantum computing. In the second part, you will learn and practice some of the basic concepts of quantum computing, including a section on mathematical tools necessary for the following weeks. In the third and final part of this week, you will be introduced to quantum algorithms and quantum computing approaches.	• Introduction & Types of Computing (25 min) • History of Classical Electronic Computing and Quantum Computing (15 min) • How Is a Quantum Computer Different? (45 min) • Quantum Gates (35 min) • Universal Quantum Algorithms (10 min) • Quantum Simulation, Emulation, and Annealing (15 min) • ✭ Check Your Understanding Questions (30 min) Check Your Understanding questions are spread throughout the week and are due at the end of the course (March 2nd) • ✭ Week 1 Quiz (30 min) Due: March 2nd @ 23:30 UTC Suggested due date to keep pace: February 8th • Key Images (3 min)
Week 2: Leading Qubit Modalities (4-5 hours) **February 9 — 15, 2026**	

In Week 2, you will learn the requirements for the physical realization of a quantum computer and for achieving quantum communication. Next, you will learn about one of the most significant obstacles in quantum computing – decoherence, or the loss of information. Then you will learn and compare the most current qubit modalities. Finally, in the case studies, you will gain more experience in the two leading modalities: Trapped ions and superconductor qubits. Here, you will not only learn more about the theory, but you will also hear scientists describe and show their laboratories.	• What Makes a Good Qubit Modality? (15 min) • Qubit Robustness (25 min) • Qubit Modalities (35 min) • ✭ Case Study: Trapped-Ion Quantum Computers (40 min) Due: March 2nd @ 23:30 UTC Suggested due date to keep pace: February 15th • ✭ Case Study: Superconducting Qubit Quantum Computers (45 min) Due: March 2nd @ 23:30 UTC Suggested due date to keep pace: February 15th • ✭ Check Your Understanding Questions (30 min) Check Your Understanding questions are spread throughout the week and are due at the end of the course (March 2nd) • ✭ Week 2 Quiz (30 min) Due: March 2nd @ 23:30 UTC Suggested due date to keep pace: February 15th • Key Images (3 min)

Week 3: Applications of Quantum Information (4-5 hours)
February 16 — 22, 2026

In the first part of this week, you will learn about quantum algorithms and the computational power of quantum computers, also called quantum advantage and quantum speed-up. In the second part, you will learn about the fundamentals of quantum communication and current communication protocols, including quantum key distribution. In the third part of this week, you will go through a case study presenting industry perspectives of quantum computation, including IBM, Google, Microsoft, QCI, IonQ, Rigetti, and D-Wave.	- Quantum Algorithms and Their Potential Speed Up (30 min) - Quantum Communication (35 min) ✭ Case Study: Industry Perspectives (30 min) Due: March 2nd@ 23:30 UTC Suggested due date to keep pace: February 22nd - Deep Dive: Introduction to Computational Complexity (35 min) ✭ Reflect and Review Activity (1.5 hrs) Written Submission due: March 2nd @ 23:30 UTC* Peer Reviews due: March 2nd @ 23:30 UTC** ✭ Check Your Understanding Questions (30 min) Check Your Understanding questions are spread throughout the week and are due at the end of the course (March 2nd) ✭ Week 3 Quiz (30 min) Due: March 2nd @ 23:30 UTC Suggested due date to keep pace: February 22nd - Key Images (3 min) *Suggested due date to keep pace: February 22nd ** Suggested due date to keep pace: February 24th
Week 4: A Simple Quantum Algorithm In Practice (4-5 hours) February 23 — March 2, 2026	

In the first part of Week 4, you will learn about the classical and quantum circuit models and compare them at a high level. Then, you will learn about the Deutsch-Josza Algorithm, with examples for one and two qubit systems. Next, you will learn about quantum software and programming tools for quantum computers. Specifically, you will be introduced to OpenQASM and its GUI interface composer. Finally, you will participate in the IBM Quantum Experience or IBM QE. Here, you will practice what you learned about OpenQASM, and use it to implement the Deutsch-Josza Algorithm in a real quantum computer.	• Circuit Models (10 min) • The Deutsch-Jozsa Quantum Algorithm (10 min) • Quantum Software (15 min) • ✭ IBM QE: QASM (40 min) Due: March 2nd @ 23:30 UTC • Lab Practicum: Deutsch-Jozsa Algorithm (2 hrs) • ✭ Check Your Understanding Questions (30 min) Check Your Understanding questions are spread throughout the week and are due at the end of the course (March 2nd) • ✭ Week 4 Quiz (30 min) Due: March 2nd @ 23:30 UTC • Key Images (3 min) • Exit Survey (10 min) • ✭ Post Assessment (10 min) Due: March 2nd @ 23:30 UTC
After the course ends...	
	March 2, 2026 • Course ends at 23:30 UTC • Discussion forums lock **March 6, 2026** • Download your Course Certificate from your student dashboard

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System Requirements & Time Zone Settings

For the best possible experience, the **latest version of Chrome or Firefox on a personal laptop or desktop computer** is recommended.

Web Browsers on Mobile Devices don't support all of features and protocols required for an optimal course experience.

Some content, like videos, are available on Mobile (iOS and Android), Safari, Internet Explorer, and MS Edge. However, these browsers don’t support the complete set of HTML5 and JavaScript required for every feature of our courseware to work.

If you are on a secured machine or network, there are likely limits to the types of content that can be downloaded, uploaded, and shared. Please be aware and abide by your local IT policies.

During Week 4's Lab Practicum/IBM Q Experience, you can complete all activities within the course platform. However, if you would like to have a more graphical IBM Experience, you can construct the circuit in an alternate site, which will be provided in the lab content, following directions described therein.

This course uses Coordinated Universal Time (UTC) for all assignment due dates. Current time in UTC:

Sat, Feb 21, 2026
16:43

To convert times to your local time zone:

- Use the following time zone converter tool

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Course Pedagogy

LEARNING OBJECTIVES

We will cover lots of ground in this course, but the main objectives are to ensure that by the end of the course, you will be able to:

- Describe one substantial difference between quantum and classical computation.
- Assess at least one potential business application for quantum computation.
- Understand engineering challenge currently faced by developers of real quantum computers.
- Evaluate one key technology requirement for quantum computers to be able to function properly.
- Utilize the IBM Quantum Experience.

PEDAGOGICAL MODEL

The design from this course follows MIT's motto *mens et manus* (mind and hands). Hands-on activities and real-world application are fundamental for acquiring significant learning.

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Grading and Completion Criteria

Please mind the due dates shown in UTC (Coordinated Universal Time) time zone. Make sure you convert them to your local time zone. Deadline extensions won't be granted.

In order to complete or pass the course, you must achieve a score of **60% or higher.** A breakdown of the grades that you earn will appear in your Progress Page.

Pre-Assessment (5% of completion percentage)

- There is one Pre-Assessment in the course that occurs at the beginning of Week 1.
- This is a required first step in order to enter the course.
- The act of completing and submitting the pre-assessment awards 5%.

Check Your Understanding (20% of completion percentage)

- Check Your Understanding questions appear in each week of the course, typically following each video.
- You have multiple chances to answer. The number of attempts is meant to ensure you are successful as often as possible.

Quizzes (40% of completion percentage)

- There are four Quizzes in the course (one per week). You have <u>one attempt</u> for each question here.
- The lowest score of the four Quizzes is dropped, so only three count towards your completion percentage.

Case Study Concept Questions (5% of completion percentage)

- Usually ~2 questions total per case study. The liberal number of attempts is meant to ensure you are successful as often as possible.

Reflect and Review Activity (10% of completion percentage)

- There is one Reflect and Review activity, in Week 3 of the course.
- In order to receive credit for this assignment, you must complete all three parts—reflection, posting, and peer reviews.

Lab Practicum/Quantum Experience (15% of completion percentage)

- There is one IBM Q Experience lab within the course, in Week 4.

Post-Assessment (5% of completion percentage)

- There is one Post-Assessment in the course that occurs at the end of Week 4.
- The act of completing and submitting the post-assessment awards 5%.

Green Check Marks

In order to get the green check mark on a course page, learners are required to attempt all assignments, including ungraded and optional assignments with submit buttons, read all content, and watch all videos to 90% completion.

Note: The green checkmarks DO NOT affect learners' grades. They are merely a tool that is supposed to be helpful in directing learners to pages in the course they have not completed. Learners should check the Progress Page at the top of the course page to view their current course score.

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Academic Integrity and Rules of Conduct

MIT takes academic honesty very seriously. The teaching team diligently monitors participant work. Should they become aware of any attempts at cheating or dishonest practices, including plagiarism, answer sharing, or improper collaborating on assignments, they reserve the right to take action, including assigning the offending participant a failing grade for the assignment; denying or revoking the MIT xPRO certificate; and/or banning the participant student from future MIT xPRO courses.

This course engages peers, instructors, and course assistants in a professional learning environment. It is important that learners are respectful towards instructors and fellow learners, and that their behaviors not interfere with nor disrupt class activities. Should any disrespectful behavior arise, it will be evaluated and appropriate measures to address the behavior will be taken.

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Certificates

Course Certificate

Upon successful completion of this course, a **Certificate of Completion** will be awarded by MIT xPRO.

Your Certificate of Completion will be posted to your MIT xPRO student dashboard the Friday after the course ends.

Participants of *Introduction to Quantum Computing* who successfully complete all course requirements are eligible to receive **2.0 Continuing Education Units** (2.0 CEUs). Your Certificate of Completion will indicate your CEU award.

CEUs are U.S. nationally-recognized means of recording non-credit/non-degree study. They are accepted by many employers, licensing agencies, and professional associations as evidence of a participant's serious commitment to the development of professional competence.

CEUs cannot be applied toward MIT undergraduate or graduate-level courses.

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